Physical Chemistry Cume
Quantum Theory
12/3/2010

The simplest quantum mechanical problem is the particle in the box: the motion of a pointlike particle with mass m between $x=0$ and $x=L$. The potential energy is zero in the box, i.e. between 0 and L. Outside the box, the potential energy is infinite and the wavefunction is zero.

(a) The exact ground-state wavefunction for this model problem is $\psi(x)=N\sin(\pi x/L)$ within the box, and zero outside. Determine the correct normalization constant N (see the integrals below), and the exact ground-state energy.

(b) If the box is 10 Å long, what is the probability that the particle is found between 0 and 3 Å?

(c) What is the mean square distance of the particle from the center of the box, i.e. the expectation value of the operator $d^2=(x - L/2)^2$? (The square root of this quantity, the rms = root-mean-square distance from the center, is often taken as the average size of the wavepacket). (see integrals below).

(d) An approximation to the ground-state wavefunction of the particle in the box is the following function: $\varphi(x) = N[Lx - x^2]$ where L is again the length of the box, and N is a normalization constant (different from the N in (a)). The box extends from 0 to L; outside of this range, the wavefunction is zero. Determine the correct normalization constant for this wavefunction.

(e) Calculate the expectation value of the kinetic energy with this approximate wavefunction, and compare it with the exact ground-state energy. Which is higher? (As the potential energy is zero, the expectation value of the kinetic energy is the same as the total energy.) Could you have predicted which will be higher, the exact energy calculated in (a) or the approximate energy calculated here?

Integrals needed:

$\int \sin(ax)^2 dx = x/2 - \sin(2ax)/(4a)$ + constant (indefinite integral)

$\int_0^L x \sin(\pi x/L)^2 \, dx = L^2/4$

$\int_0^L x^2 \sin(\pi x/L)^2 \, dx = L^3(2\pi^2-3)/(12\pi^2)$